Project Report

Predictive Analysis for product demand

SARIMA - LSTM

Using times-series forecasting models to estimate product demand on lens frames for a distributor company.

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View Code on Github

Abstract

This document presents the application of statistical and machine learning methods for predicting the sales of lens frames of a distributor company in Mexico. The project's goal was to generate product demand forecasts two years into the future by grouping products by brand and using the information as input for an future Supply Chain Management strategy (SCM). Time-series forecasting models, specifically SARIMA and LSTM neural networks, were employed and their performance capture the sales trends and patterns was evaluated and compared.
This report provides a comprehensive documentation of the entire development and decision-making processes involved in the project.

Keywords: times-series, forecasting, SARIMA, LSTM, neural networks, machine learning, data science

Resumen

En este informe se describe la aplicación de métodos estadísticos y de aprendizaje automático para predecir la demanda de productos, específicamente armazones para lentes de una empresa distribuidora mexicana. El objetivo del proyecto fue generar pronósticos de ventas a dos años en el futuro, agrupando los productos por marca para utilizar la información en un modelo de planeación de inventario. Para lograr esto, se utilizaron el modelo SARIMA y una red neuronal recurrente de tipo LSTM, y se evaluó y comparó su desempeño en describir los patrones y tendencias de ventas para cada una de las marcas analizadas. Este informe proporciona una documentación detallada de todo el proceso de desarrollo y toma de decisiones para cumplir con el objetivo establecido.

Palabras clave: series de tiempo, pronóstico, SARIMA, LSTM, red neuronal, aprendizaje automático, ciencia de datos

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1. User story

This project was completed as a freelancing job. The team in charge of acquiring stock for the distributor company, required an accurate demand forecast, necessary to determine the appropriate amount of product to purchase in order to meet customer demand and ensure that their stock would not run out at any time.

2. Background

Predictive analysis has huge potential on Supply Chain Management, as it allows companies to make informed decisions on inventory management and efficiently plan their re-stocking strategies.

As the goal is to produce an accurate and robust forecast, two powerful forecasting methods are applied and compared: SARIMA and LSTM.

SARIMA stands for *Seasonal Autoregressive Integrated Moving Average*. The SARIMA model is designed to forecast future values of a time series by analyzing its historical data, and has the ability to learn about the data properties such as trend, autocorrelation, and variations like seasonality and periodicity. It is particularly effective in analyzing time series data that has clear periodic patterns, such as monthly or quarterly sales figures. For this reason, SARIMA was chosen to evaluate the sales data in this project, as it is grouped by month.

On the other hand, LSTM models are a highly effective type of recurrent neural network that are well-suited for analyzing sequential data, such as time series. As neural networks, they have the ability to discern patterns within data. However, the primary advantage of LSTM models is their capacity to accurately capture long-term dependencies in a series, such as trends, which makes them particularly useful for forecasting where historical data extends over a significant period with numerous observations recorded, and where the preservation of all relevant information is of critical importance.

3. Data

The dataset utilized in this project was obtained from the distributor company and included actual sales figures recorded on a monthly basis over a period of 48 months.

3.1 Data preparation

Initially, the data consisted of raw sales and return records, which required merging to determine the company's total sales. Sales data were grouped by brand, rather than individual SKU, and only the brand, date, and quantity of each registered sale were retained for analysis purposes.

Here is a fragment of the utilized dataset:

BRAND	DATE	QTY	MONTH
BANDERAS	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
NANO VISTA	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
AXESSONE	2023-01-02	1	1
AXESSONE	2023-01-02	1	1
BANDERAS	2023-01-02	1	1
AXESSCLIP	2023-01-02	1	1

Figure 1: Fragment of used dataset

3.2 Initial data analysis

First, to gain insight into the behavior of the data, STL decomposition was conducted on all the entire dataset registers, as well as on individual subsets segmented by brand.

The resulting plots reveal that the trend component plays a more dominant role in shaping the data behavior than the seasonal and residual components. This underscores the significance of the trend component in capturing the overall pattern of the data.

4. Approach

Describe the techniques that you will use for managing and analyzing your data and the tool(s) that you will use for data management/data analysis and evaluation. Motivate why you have chosen a certain method over others and describe how you plan to evaluate your solution. This section should give a detailed account of what you did in order to answer your questions.

5. Results

Describe your final data analysis results and everything that you discovered on the way the put you forward and you think is worth describing here. Elaborate on the significance of your findings/results and convey your thoughts.

6. Discussion

What do the results mean (their significance and to whom) and how do they answer your research questions/hypotheses? From the evaluation and validation of the tool, what can you conclude?

6.1 Limitations and Challenges

What could have been investigated if given more time? What have been difficult when solving the problem and getting answers for your research questions/hypotheses?

7. Conclusion

Write the conclusion. What did you gain from the project assignment? Briefly explain your questions/hypotheses, findings, and meaningful discussion points in relation to the data collection, data management, data analysis, visualization and interaction concepts, and evaluation of your tool. What additional investigations need to be performed (or what is the limitation) in order to say that your solution is a good one for the problem?

8. Reflections on own work

- Describe how you decided to scope (and/or re-scope) your problem formulation during the work, and given the data you had access to.
- Describe how you searched for knowledge on how to scope the DS question you wanted to answer
- understand how to implement, test, and validate your results.
- Which sources helped you to get progress and how?
- What would you have done differently if you were to start over again, for understanding the problem faster and for understanding what could be done with the data you had access to?
- What else is there that you would have changed about this assignment?

A. Appendices

If your paper includes appendices, they are to be placed at the very end of your paper. If you have conducted a survey or interviews, the questionnaire and interview guide are to be included as appendices. If you have made some ML implementation, attach the code. If you are unsure about what to include as an appendix, consult with your lecturer or supervisor. Make sure to name the appendices correctly, like A, B and C.

A.1 LaTeX usage examples

This chapter is included solely as a help and reference for using LaTeX. It should not appear in the final document.

A.2 Introduction

This chapter shows examples of using LaTeXfor common operations.

A.3 Styles

Styles such as **bold**, *italic*, and <u>underline</u> can be applied to text. You can also use apply colors, and <u>*combine*</u> **styles**. It is recommended to use only bold to emphasize, and not abuse this resource.

A.4 Listings

With itemize you can create unnumbered lists:

- strawberries
- Peaches
- Pineapples
- Nectarines

Similarly, enumerate allows you to create numbered lists:

1. Prepare the memory of the TFG
2. Prepare the presentation
3. Present the TFG
4. Apply for the Bachelor's degree

A.5 Subsections

Subsections can be defined with the subsection command:

A.5.1 First subsection

This is a subsection

A.5.2 Second subsection

This is another subsection.

A.6 Images and figures

All images and figures in the document will be placed in the “fig” folder. They can be included as follows:

Figure 2: An example of image

Note that the figures are automatically numbered according to the chapter and the number of figures that have previously appeared in that chapter. There are many ways to define the size of a figure, but it is advisable to use the one shown in this example: the width of the figure is defined as a percentage of the total width of the page, and the height is automatically scaled. In this way, the maximum width of a figure would be 1.0 * textwidth, which would ensure that it is displayed at the maximum possible size without exceeding the margins of the document.

Note that LaTeX tries to include figures in the same place where they are declared, but sometimes this is not possible due to space constraints. In those cases, LaTeX will place the figure as close to its declaration as possible, perhaps on a different page. This is normal behavior and should not be avoided.

A.7 References

Notice how the “label” command has been used several times in the source code for this section. This command allows you to mark an element, be it a chapter, section, figure, etc. to make a numeric reference to it. To reference a “label”, use the “ref” command including the name of the reference:

This is the chapter 2.

Examples of styles are shown in the A.3 section.

The subsection A.5.1 explains...

In Figure 2 we see that...

This saves us from having to directly write the indexes of the sections and figures we want to mention, since LaTeX does it for us and also takes care of keeping them updated in case they change (try moving this chapter to the end of the document and see how all referenced indices are updated automatically). Also, "ref" references act as hyperlinks within the document that take you to the referenced element when you click on them.

It is usual to name "label" with a prefix indicating the type of element to find it later more easily, but it is not mandatory.

A.8 Code snippets

Code snippets can be included via listing:

```
num = float(input("Enter␣a␣number:␣"))
if num > 0:
   print("Positive␣number")
elif num == 0:
   print("Zero")
else:
   print("Negative␣number")
```

Code extraction 1: Python code

A wide variety of languages are supported:

```
public class Test {
    public static void main(String[] args) {
        System.out.println("Hello,␣world!");
    }
}
```

Code extraction 2: Java coode

Code snippets can also be referenced via label/ref: Code snippets 1 and 2.

A.9 Links

You can link to an external website using the url command: https://www.example.com. A link can also be linked to text using the href command: example domain.

A.10 Citations and bibliography

In LaTeX, bibliography items are stored in a bibliographic file in a format called BibTeX, in the case of this project they are in "bibliography.bib". To cite an element, use

the “cite” command. You can cite both scientific papers [1] or books [3] as well as web links [2]. Citations are automatically numbered and included in the bibliography section of the document.

Note how bibliographic items stored in “bibliography.bib” have an associated tag, which is the one included when citing them using cite. Adding a reference to the bibliographic file does not make it automatically appear in the bibliography section of the work, it is necessary to cite it somewhere in it.

A.11 Equations

LaTeX has a powerful engine for displaying mathematical equations and an extensive catalog of mathematical symbols. The math environment can be activated in many ways. To include simple equations in a text, they can be surrounded by dollar signs: $1 + 2 = 3$, $\sqrt{81} = 3^2 = 9$, $\forall x \in y \; \exists \; z : S_z < 4$.

More complex equations can be expressed separately and are numbered: equation 1.

$$\lim_{x \to 0} \frac{e^x - 1}{2x} \overset{\left[\frac{0}{0}\right]}{\underset{\mathrm{H}}{=}} \lim_{x \to 0} \frac{e^x}{2} = \frac{1}{2} + 7 \int_0^2 \left(-\frac{1}{4} \left(e^{-4t_1} + e^{4t_1 - 8} \right) \right) dt_1 \tag{1}$$

There is here an extensive list of symbols that can be used in math mode.

A.12 Special characters and symbols

Some characters and symbols must be escaped in order to be rendered in the document, as they have a special meaning in LaTeX. Some of them are:

- The dollar sign $ is used for equations.
- The percentage % is used for comments in the source code.
- The euro symbol € often causes problems if typed directly.
- The underscore _ is used for subscripts in math mode.
- Quotes must be expressed ‘like this’ for single quotes and “like this” for double quotes. Spanish quotation marks can be expressed “like this”.
- The backslash \ is used for LaTeX commands.
- Other symbols that must be escaped include the braces { }, the ampersand &, the hash #, and the greater-than > and less-than < symbols.

Bibliography

[1] Tim Berners-Lee, Robert Cailliau, Ari Luotonen, Henrik Frystyk Nielsen, and Arthur Secret. The world-wide web. *Communications of the ACM*, 37(8):76–82, 1994.

[2] University of Skövde. Regulations, forms and templates, 2022. URL https://www.his.se/en/research/doktorandhandbok/regulations-forms-and-templates/.

[3] John M Swales and Christine B Feak. *Academic writing for graduate students*. University of Michigan Press Ann Arbor, MI, 1994.